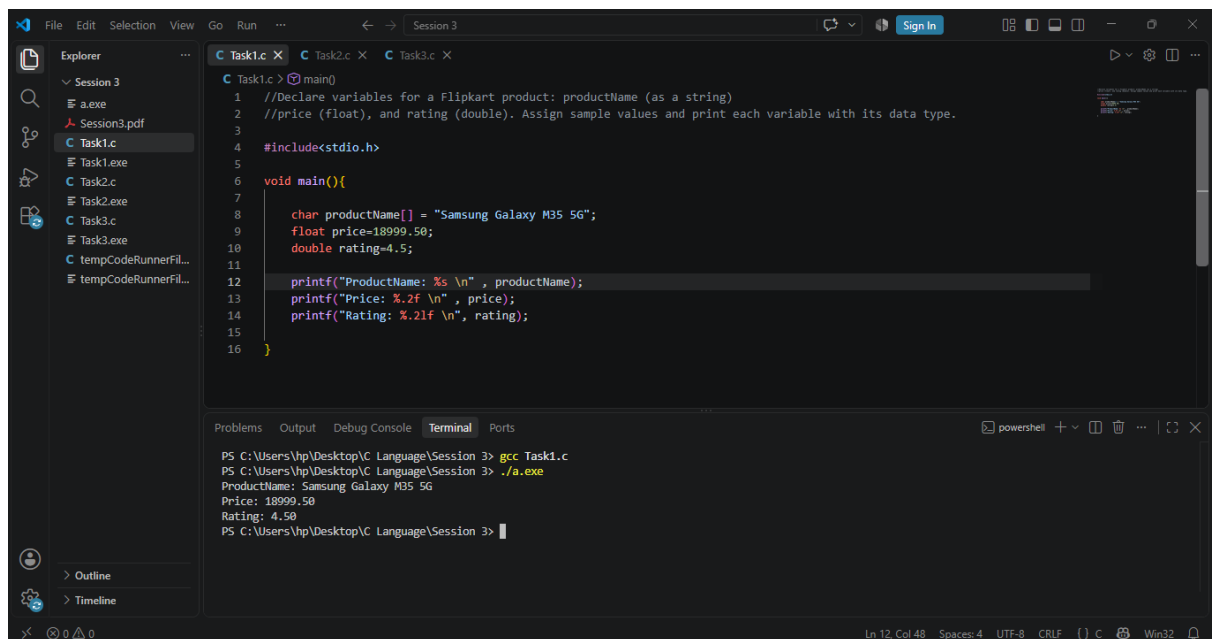


1. Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.



The screenshot shows the Visual Studio Code interface with a C program in a file named `Task1.c`. The program declares three variables: `productName` (a string), `price` (a float), and `rating` (a double). It assigns sample values to these variables and prints them using `printf`. The terminal output shows the execution of the program, displaying the product name, price, and rating.

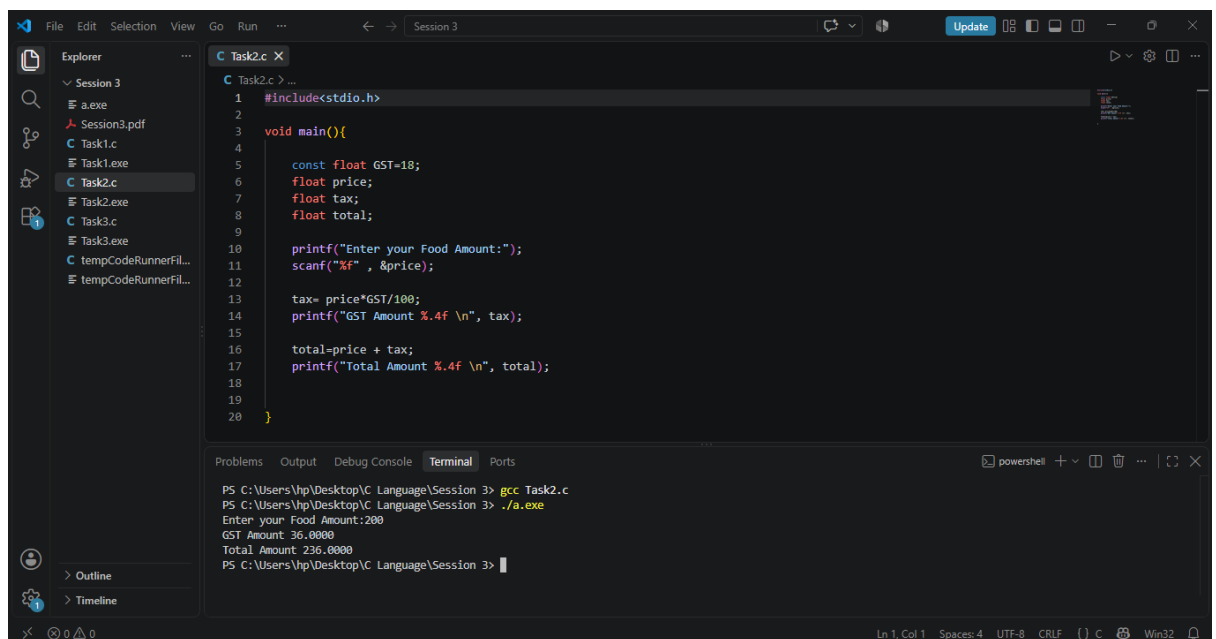
```
1 //Declare variables for a Flipkart product: productName (as a string)
2 //price (float), and rating (double). Assign sample values and print each variable with its data type.
3
4 #include<stdio.h>
5
6 void main(){
7
8     char productName[] = "Samsung Galaxy M35 5G";
9     float price=18999.50;
10    double rating=4.5;
11
12    printf("ProductName: %s \n", productName);
13    printf("Price: %.2f \n", price);
14    printf("Rating: %.2lf \n", rating);
15
16 }
```

Terminal Output:

```
PS C:\Users\hp\Desktop\C Language\Session 3> gcc Task1.c
PS C:\Users\hp\Desktop\C Language\Session 3> ./a.exe
ProductName: Samsung Galaxy M35 5G
Price: 18999.50
Rating: 4.50
PS C:\Users\hp\Desktop\C Language\Session 3>
```

2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.



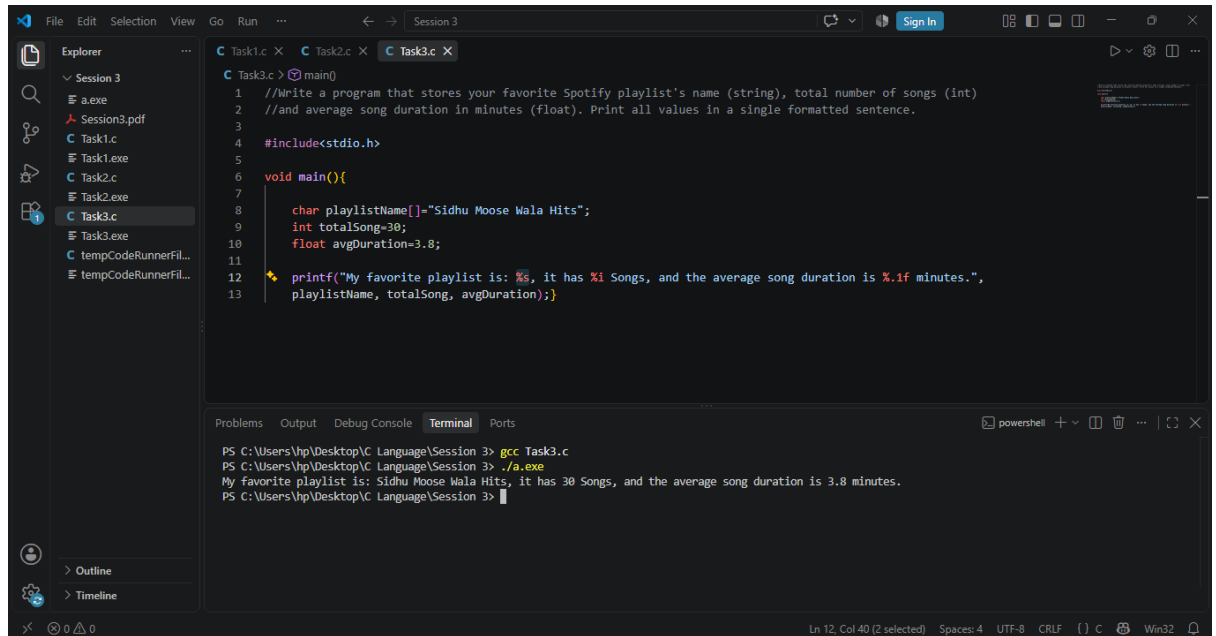
The screenshot shows the Visual Studio Code editor with a C program named Task2.c. The program includes `<stdio.h>` and defines a `main` function. Inside `main`, a constant float `GST` is set to 18. It prompts the user for a food amount, calculates the GST (18% of the food amount), and then calculates the total amount (food amount plus GST). The program uses `printf` for output and `scanf` for input.

```
1 #include<stdio.h>
2
3 void main(){
4
5     const float GST=18;
6     float price;
7     float tax;
8     float total;
9
10    printf("Enter your Food Amount:");
11    scanf("%f", &price);
12
13    tax= price*GST/100;
14    printf("GST Amount %.4f \n", tax);
15
16    total=price + tax;
17    printf("Total Amount %.4f \n", total);
18
19
20 }
```

The terminal window shows the execution of the program. It prompts for the food amount, which is entered as 200. The output shows a GST amount of 36.0000 and a total amount of 236.0000.

```
PS C:\Users\hp\Desktop\C Language\Session 3> gcc Task2.c
PS C:\Users\hp\Desktop\C Language\Session 3> ./a.exe
Enter your Food Amount:200
GST Amount 36.0000
Total Amount 236.0000
PS C:\Users\hp\Desktop\C Language\Session 3>
```

3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.



The screenshot shows the Visual Studio Code interface with a C program in `Task3.c`. The program stores a playlist name, total songs, and average duration, then prints them in a formatted sentence. The terminal shows the compilation and execution of the program.

```
1 //Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int)
2 //and average song duration in minutes (float). Print all values in a single formatted sentence.
3
4 #include<stdio.h>
5
6 void main(){
7
8     char playlistName[]="Sidhu Moose Wala Hits";
9     int totalSong=30;
10    float avgDuration=3.8;
11
12    printf("My favorite playlist is: %s, it has %i Songs, and the average song duration is %.1f minutes.",
13          playlistName, totalSong, avgDuration);
```

Terminal Output:

```
PS C:\Users\hp\Desktop\C Language\Session 3> gcc Task3.c
PS C:\Users\hp\Desktop\C Language\Session 3> ./a.exe
My favorite playlist is: Sidhu Moose Wala Hits, it has 30 Songs, and the average song duration is 3.8 minutes.
PS C:\Users\hp\Desktop\C Language\Session 3>
```

4. Identify and correct the invalid variable names from this list: 1stPlayer, player_age, \$score, total-marks, userName.

Hint: Apply standard variable naming rules for your programming language.

Given Variable	Status	Correct Variable Name
1stPlayer	Invalid	firstPlayer,Player1
player_age	Valid	Player_age
\$score	Invalid	Score
userName	Valid	userName